

SBC 201 Chapter 20 Aluminum — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–5 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 20, source file Reference\SBC 201 2024\source_reference\Chapter_20 – ALUMINUM.txt.
- **Extraction audit:** Skill-finetune run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, building-language triggers, named exceptions, check-specific actions). Internal inventory: **0** independently checkable numeric records (**0** Verified, **0** Verify source). The attached extract has no appended tables and no OCR hold points.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, structural design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from AA ADM 1, AA ASM 35, Chapter 16, SBC 501, SBC 701, commentary examples, bibliography edition years, or the existing chapter summary. Where Chapter 20 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection, mixed-use podium, storey count and structural system are unconfirmed. Chapter 20 does not branch on NFPA 13 versus 13R or on occupancy group.
4. Charging **2002.1** names **AA ASM 35** and **AA ADM 1** with **no edition year**. Bibliography names ADM1-15 (2015) and ASM 35-00 (2000). Neither commentary year is adopted as a charging substitution.
5. General Comments state that specialty products (storefront framing, architectural hardware) are outside this chapter and that HVAC/plumbing aluminum is addressed in SBC 501 / SBC 701. Those commentary statements are not treated as numbered code requirements and no 501/701 values are imported.
6. Typical tower aluminum (windows, curtain wall, railings, composite panels) is not given millimetre geometry in this chapter. Only aluminum used **for structural purposes** is charged to the named AA standards, with nominal loads from Chapter 16.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, use, occupant load, or mixed-use/podium condition exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 20 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 20 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 20 application	Required project action
Structural aluminum present	Unconfirmed: whether any load-carrying members, flashings or sheet roofing are aluminum	2002.1 charges only aluminum used for structural purposes	Structural engineer to flag every aluminum member, flashing and sheet-roofing element that carries load or weathers the enclosure
AA ADM 1 / ASM 35 edition	Charging 2002.1 has no year; bibliography names 2015 and 2000	Controls allowable stresses, LRFD factors, sheet-metal work and flashing details	AHJ/code consultant to lock the referenced AA editions; do not silently adopt commentary years
Chapter 16 loads	Unconfirmed; values not in this chapter	2002.1 requires nominal loads to be the Chapter 16 minima	Use the structural design criteria from Chapter 16; do not invent wind, seismic or live-load figures here
Architectural aluminum	Typical tower will have windows, curtain wall, railings and panels	This chapter does not publish millimetre, thickness or hardware rules for those products	Detail architectural aluminum from the product specification and the structural engineer of record — not from Chapter 20 commentary
Sprinkler / mixed use / storeys	Unconfirmed	Chapter 20 has no occupancy, sprinkler or storey branch	Do not wait on NFPA 13 vs 13R to apply 2001–2002; those decisions do not change this chapter
NOC / structural PE	Unconfirmed	AA ADM 1 design and Chapter 16 load takeoff cannot be concluded from this extract	Engage the structural engineer of record before freezing aluminum member sizes

4. Scope and charging standards

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2001.1	Chapter aluminum scope	This chapter governs quality, design, fabrication and erection of aluminum	Aluminum quality, design, fabrication or erection on the building	None stated	Direct	List aluminum work packages on the drawing index and send structural aluminum through Sections 2001–2002	Verified
2002.1	Structural aluminum standards	Aluminum used for structural purposes shall comply with AA ASM 35 and AA ADM 1	Structural aluminum members and elements	None stated	External verification	Name both AA standards on structural aluminum drawings and specifications; do not import stresses, thicknesses or connection values from those manuals into this matrix	Verified
2002.1	Structural aluminum design loads	Nominal loads shall be the minimum design loads required by Chapter 16	Structural aluminum designed under this chapter	None stated	External verification	Take aluminum member loads from the Chapter 16 structural criteria; do not copy wind, seismic or live-load figures here	Verified

5. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger is confirmed (structural aluminum actually present).
2. There are no **Verify source** rows in this extract. The attached Chapter 20 text is clean numbered code with no appended tables.
3. Do not import AA ADM 1 stresses, AA ASM 35 flashing thicknesses, or Chapter 16 load values into issued drawings from this matrix.
4. Do not adopt bibliography years (ADM1-15 / ASM 35-00) as the charging edition without AHJ confirmation.
5. Do not treat General Comments on storefront, architectural hardware, SBC 501 or SBC 701 as numbered code cells, and do not import values from those parts.
6. Record structural-aluminum and AA-edition decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

6. Coverage summary

Internal inventory of the attached Chapter 20 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 0
- **Verified:** 0
- **Verify source:** 0

Counts by top-level section

Top-level section	Records
2001	0
2002	0

No appended tables in the attached extract.

Coverage cross-check against SBC 201 Chapter 20 Aluminum (2024)_CS.md was topics-only: chapter scope; structural aluminum charged to AA ASM 35 and AA ADM 1; Chapter 16 loads; commentary notes on specialty products and SBC 501/701. No CS.md value was copied into a matrix cell.

7. Unresolved-source register

No OCR, flattened-table, page-split or footnote-attachment hold points. The attached extract contains two unambiguous code paragraphs and no tables.

Affected table / clause	Why unverified	Control note
—	None	No Verify source inventory records